

Session 7: Analysis of Variance (ANOVA) and Non-Parametric Tests

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One-Way Analysis of Variance (One-Way ANOVA)

One-Way Analysis of Variance (ANOVA) is a statistical test used to determine whether there are significant differences among the means of three or more independent groups based on a single categorical factor.

Purpose

To test whether at least one group mean differs significantly from the others.

Hypotheses

- **Null Hypothesis (H_0):** All group means are equal.

$$[H_0: \mu_1 = \mu_2 = \dots = \mu_k]$$

- **Alternative Hypothesis (H_1):** At least one group mean is different.

Example

A researcher wants to compare the average malaria prevalence among three regions:

- Region A
- Region B
- Region C

One-Way ANOVA can determine whether the mean prevalence differs across these regions.

Assumptions

1. Observations are independent.
2. The dependent variable is continuous.
3. The dependent variable is approximately normally distributed within each group.
4. Homogeneity of variances across groups.

ANOVA Test Statistic

$$[F = \frac{\text{Mean Square Between Groups}}{\text{Mean Square Within Groups}}]$$

A large F-statistic suggests greater differences among group means than would be expected by chance.

Decision Rule

- If **p-value** < α (e.g., 0.05), reject H_0 .
- If **p-value** $\geq \alpha$, fail to reject H_0 .

Interpretation

- **Significant result:** At least one group mean differs.
- **Non-significant result:** No evidence of differences among group means.

Follow-Up Analysis

If ANOVA is significant, perform a **post hoc test** (e.g., Tukey's HSD Test) to identify which specific groups differ.

Reporting Example

A one-way ANOVA was conducted to compare mean malaria prevalence across three regions. The analysis revealed a statistically significant difference among the group means, $F(2, 27) = 5.43$, $p = 0.010$.

Import the packages

```
In [1]: import pandas as pd
from statsmodels.stats.weightstats import ttest_ind
import numpy as np
import pingouin as pg
df = pd.read_csv("german_credit__data.csv")
```

Load from local CSV or URL

```
url = "https://raw.githubusercontent.com/selva86/datasets/master/GermanCredit.csv"
```

```
df = pd.read_csv(url)
```

Removing missing values

```
In [2]: df = df.dropna()
#df
```

```
In [3]: import pandas as pd
from scipy.stats import f_oneway
import numpy as np
from scipy.stats import f_oneway
from statsmodels.stats.multicomp import pairwise_tukeyhsd
```

```
In [4]: # Perform one-way ANOVA
f_statistic, p_value = f_oneway(*[df[df['Housing'] == category]['Duration'] for cat

# Print results
print("F-statistic:", f_statistic)
print("p-value:", p_value)
```

F-statistic: 10.57660878539277
p-value: 3.14612746390347e-05

Perform One-Way ANOVA: Alternative Method

```
In [5]: anova_results = pg.anova(dv='Duration', between='Housing', data=df, detailed=True)

# Print Results
anova_results
```

```
Out[5]:
```

	Source	SS	DF	MS	F	p-unc	np2
0	Housing	3174.784070	2	1587.392035	10.576609	0.000031	0.039162
1	Within	77894.198689	519	150.085161	NaN	NaN	NaN

Tukey Honest Significant Difference

```
In [6]: tukey_results = pg.pairwise_tukey(dv='Duration', between='Housing', data=df)

# Print results
tukey_results
```

```
Out[6]:
```

	A	B	mean(A)	mean(B)	diff	se	T	p-tukey	hedges
0	free	own	27.861538	20.30659	7.554948	1.655006	4.564906	0.000019	0.601890
1	free	rent	27.861538	20.75000	7.111538	1.923195	3.697773	0.000703	0.561412
2	own	rent	20.306590	20.75000	-0.443410	1.348970	-0.328702	0.942190	-0.037350

Explain the output

Two-Way Analysis of Variance

Generate the Data set

```
In [7]: import pandas as pd
import pingouin as pg
import numpy as np
```

```
In [8]: # Sample Data (Replace with your actual dataset)

np.random.seed(42) # For reproducibility
df = pd.DataFrame({
    'Sponsorship': np.random.choice(['Scholarship', 'Self-funded'], size=30),
    'Gender': np.random.choice(['Male', 'Female'], size=30),
    'GPA': np.random.randn(30) * 0.5 + 3.0 # Simulated GPA values
})
#df.head()
```

Convert categorical variables to factors

```
In [9]: df['Sponsorship'] = df['Sponsorship'].astype('category')
df['Gender'] = df['Gender'].astype('category')
```

Perform Two-Way ANOVA

```
In [10]: anova_results = pg.anova(dv='GPA', between=['Sponsorship', 'Gender'], data=df, deta
```

Print results

```
In [11]: #anova_results
```

Perform Tukey's HSD Test

```
In [12]: df['Group'] = df['Sponsorship'].astype(str) + " - " + df['Gender'].astype(str) # C
tukey = pairwise_tukeyhsd(df['GPA'], df['Group'])

# Print Tukey results
#print(tukey)
```

Use German Credit for Two way ANOVA

```
In [13]: df = pd.read_csv("german_credit_data.csv")
#df.head()
```

```
In [14]: df = df.dropna()
#df.head()
```

Convert the Variables of Interest into Category Variable

```
In [15]: df['Housing'] = df['Housing'].astype('category')
df['Purpose'] = df['Purpose'].astype('category')
```

View the Frequency Table for the Categorical Variable

Frequency Table for 'Housing'

```
In [16]: import pandas as pd

housing_freq = df['Housing'].value_counts().reset_index()
housing_freq.columns = ['Category', 'Frequency']
housing_freq['Percentage'] = (housing_freq['Frequency'] / housing_freq['Frequency'])
#housing_freq
```

Frequency Table for 'Purpose'

```
In [17]: import pandas as pd

purpose_freq = df['Purpose'].value_counts().reset_index()
purpose_freq.columns = ['Category', 'Frequency']
purpose_freq['Percentage'] = (purpose_freq['Frequency'] / purpose_freq['Frequency'])
#purpose_freq
```

Crosstabulation for Housing vs Purpose

```
In [18]: crosstab = pd.crosstab(df['Housing'], df['Purpose'], margins=True, margins_name="Total")
#crosstab
```

Convert Crosstab to DataFrame (Optional)

```
In [19]: crosstab_df = crosstab.reset_index()
#crosstab_df
```

Grouped Summary Statistics

```
In [20]: import pandas as pd

# Grouped summary statistics (Credit amount across Housing type)
summary_stats = df.groupby('Housing')['Credit amount'].describe()

# Display results
summary_stats
```

C:\Users\Admin\AppData\Local\Temp\ipykernel_12280\3242974759.py:4: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.

```
summary_stats = df.groupby('Housing')['Credit amount'].describe()
```

```
Out[20]:
```

	count	mean	std	min	25%	50%	75%	max
Housing								
free	65.0	4846.384615	3583.997767	781.0	1872.00	3844.0	6615.0	14782.0
own	349.0	3003.504298	2770.028290	338.0	1225.00	2136.0	3857.0	18424.0
rent	108.0	3224.694444	2719.530785	276.0	1439.75	2458.5	3636.5	15945.0

```
In [21]: import warnings
warnings.simplefilter(action='ignore')

# Grouped summary statistics (Credit amount across Purpose)
summary_stats = df.groupby('Purpose')['Credit amount'].describe()

# Display results
#summary_stats
```

Grouped by Housing and Purpose

```
In [22]: import pandas as pd
import warnings
warnings.simplefilter(action='ignore')

# Grouped summary statistics for Housing and Purpose
summary_stats = df.groupby(['Housing', 'Purpose'])['Credit amount'].describe()

# Display results
#summary_stats
```

Perform the Two-Way ANOVA

```
In [23]: import warnings
warnings.simplefilter(action='ignore')

anova_results = pg.anova(dv='Credit amount', between=['Housing', 'Purpose'], data=d
```

Print the Results

```
In [24]: #anova_results
```

Tukey Honesty Significant Difference

```
In [25]: df['Group'] = df['Housing'].astype(str) + " - " + df['Purpose'].astype(str) # Comb
tukey = pairwise_tukeyhsd(df['Credit amount'], df['Group'])
```

```
In [26]: # Print Tukey results
print(tukey)
```

Multiple Comparison of Means - Tukey HSD, FWER=0.05

=====					
=====					
upper	group1 reject	group2	meandiff	p-adj	lower

	free - business	free - car	475.3143	1.0	-5342.89
6293.5186	False				
	free - business	free - education	609.25	1.0	-5938.356
7156.856	False				
	free - business	free - furniture/equipment	-285.5556	1.0	-6733.1922
6162.0811	False				
	free - business	free - radio/TV	-2608.0	0.9989	-9446.7514
4230.7514	False				
	free - business	free - repairs	-3515.0	0.9999	-14682.6343
7652.6343	False				
	free - business	free - vacation/others	3137.6667	0.998	-4759.0433
11034.3766	False				
	free - business	own - business	-979.0263	1.0	-6779.0707
4821.0181	False				
	free - business	own - car	-1584.5146	1.0	-7249.0658
4080.0367	False				
	free - business	own - domestic appliances	-3371.5	0.9739	-10210.2514
3467.2514	False				
	free - business	own - education	-2079.9231	0.9998	-8274.612
4114.7659	False				
	free - business	own - furniture/equipment	-1673.9	1.0	-7376.1151
4028.3151	False				
	free - business	own - radio/TV	-2397.3861	0.9956	-8063.5245
3268.7522	False				
	free - business	own - repairs	-1711.5	1.0	-7954.3974
4531.3974	False				
	free - business	own - vacation/others	5616.8333	0.2824	-1221.9181
12455.5848	False				
	free - business	rent - business	1475.8333	1.0	-4767.064
7718.7307	False				
	free - business	rent - car	-1306.7143	1.0	-7124.9186
4511.49	False				
	free - business	rent - education	-2077.1429	0.9999	-8751.0809
4596.7952	False				
	free - business	rent - furniture/equipment	-1814.7143	1.0	-7690.0545
4060.6259	False				
	free - business	rent - radio/TV	-2567.0	0.9939	-8476.3566
3342.3566	False				
	free - business	rent - repairs	-2321.0	1.0	-13488.6343
8846.6343	False				
	free - car	free - education	133.9357	1.0	-3656.1325
3924.0039	False				
	free - car	free - furniture/equipment	-760.8698	1.0	-4375.4918
2853.7521	False				
	free - car	free - radio/TV	-3083.3143	0.5391	-7356.7197
1190.0911	False				
	free - car	free - repairs	-3990.3143	0.9973	-13798.9599
5818.3314	False				
	free - car	free - vacation/others	2662.3524	0.9886	-3155.8519

8480.5567	False	free - car	own - business	-1454.3406	0.7539	-3720.1711
811.4899	False	free - car	own - car	-2059.8288	0.0168	-3952.0798
-167.5779	True	free - car	own - domestic appliances	-3846.8143	0.1431	-8120.2197
426.5911	False	free - car	own - education	-2555.2374	0.2998	-5696.518
586.0433	False	free - car	own - furniture/equipment	-2149.2143	0.0204	-4151.3957
-147.0329	True	free - car	own - radio/TV	-2872.7004	0.0	-4769.6971
-975.7038	True	free - car	own - repairs	-2186.8143	0.6661	-5422.1258
1048.4972	False	free - car	own - vacation/others	5141.519	0.0034	868.1136
9414.9245	True	free - car	rent - business	1000.519	0.9999	-2234.7924
4235.8305	False	free - car	rent - car	-1782.0286	0.4041	-4093.9485
529.8914	False	free - car	rent - education	-2552.4571	0.7646	-6556.82
1451.9057	False	free - car	rent - furniture/equipment	-2290.0286	0.1026	-4742.19
162.1328	False	free - car	rent - radio/TV	-3042.3143	0.0035	-5574.8957
-509.7329	True	free - car	rent - repairs	-2796.3143	1.0	-12604.9599
7012.3314	False	free - education	free - furniture/equipment	-894.8056	1.0	-5594.288
3804.6769	False	free - education	free - radio/TV	-3217.25	0.8131	-8440.4327
2005.9327	False	free - education	free - repairs	-4124.25	0.9977	-14382.3772
6133.8772	False	free - education	free - vacation/others	2528.4167	0.9987	-4019.1893
9076.0227	False	free - education	own - business	-1588.2763	0.9957	-5350.4075
2173.8549	False	free - education	own - car	-2193.7646	0.8087	-5743.4486
1355.9195	False	free - education	own - domestic appliances	-3980.75	0.427	-9203.9327
1242.4327	False	free - education	own - education	-2689.1731	0.8071	-7035.1238
1656.7776	False	free - education	own - furniture/equipment	-2283.15	0.7761	-5892.6339
1326.3339	False	free - education	own - radio/TV	-3006.6361	0.231	-6558.8523
545.58	False	free - education	own - repairs	-2320.75	0.95	-6735.1451
2093.6451	False	free - education	own - vacation/others	5007.5833	0.0791	-215.5994
10230.766	False	free - education	rent - business	866.5833	1.0	-3547.8117
5280.9784	False	free - education	rent - car	-1915.9643	0.9661	-5706.0325

1874.1039	False					
	free - education	rent - education	-2686.3929	0.9393	-7691.8464	
2319.0607	False					
	free - education	rent - furniture/equipment	-2423.9643	0.7929	-6301.1719	
1453.2433	False					
	free - education	rent - radio/TV	-3176.25	0.3111	-7104.8136	
752.3136	False					
	free - education	rent - repairs	-2930.25	1.0	-13188.3772	
7327.8772	False					
	free - furniture/equipment	free - radio/TV	-2322.4444	0.9891	-7419.7488	
2774.8599	False					
	free - furniture/equipment	free - repairs	-3229.4444	0.9999	-13424.0532	
6965.1643	False					
	free - furniture/equipment	free - vacation/others	3423.2222	0.9451	-3024.4145	
9870.8589	False					
	free - furniture/equipment	own - business	-693.4708	1.0	-4278.7889	
2891.8473	False					
	free - furniture/equipment	own - car	-1298.959	0.9987	-4660.6745	
2062.7565	False					
	free - furniture/equipment	own - domestic appliances	-3085.9444	0.8349	-8183.2488	
2011.3599	False					
	free - furniture/equipment	own - education	-1794.3675	0.9949	-5988.1918	
2399.4567	False					
	free - furniture/equipment	own - furniture/equipment	-1388.3444	0.9974	-4813.1434	
2036.4546	False					
	free - furniture/equipment	own - radio/TV	-2111.8306	0.7873	-5476.2197	
1252.5585	False					
	free - furniture/equipment	own - repairs	-1425.9444	0.9998	-5690.6552	
2838.7664	False					
	free - furniture/equipment	own - vacation/others	5902.3889	0.0065	805.0845	
10999.6933	True					
	free - furniture/equipment	rent - business	1761.3889	0.9967	-2503.3219	
6026.0997	False					
	free - furniture/equipment	rent - car	-1021.1587	1.0	-4635.7807	
2593.4632	False					
	free - furniture/equipment	rent - education	-1791.5873	0.9993	-6665.5425	
3082.3679	False					
	free - furniture/equipment	rent - furniture/equipment	-1529.1587	0.9968	-5235.0477	
2176.7303	False					
	free - furniture/equipment	rent - radio/TV	-2281.4444	0.8321	-6041.0304	
1478.1415	False					
	free - furniture/equipment	rent - repairs	-2035.4444	1.0	-12230.0532	
8159.1643	False					
	free - radio/TV	free - repairs	-907.0	1.0	-11353.3654	
9539.3654	False					
	free - radio/TV	free - vacation/others	5745.6667	0.2432	-1093.0848	
12584.4181	False					
	free - radio/TV	own - business	1628.9737	0.9988	-2619.6741	
5877.6215	False					
	free - radio/TV	own - car	1023.4854	1.0	-3038.2425	
5085.2134	False					
	free - radio/TV	own - domestic appliances	-763.5	1.0	-6347.3172	
4820.3172	False					
	free - radio/TV	own - education	528.0769	1.0	-4245.2515	
5301.4053	False					
	free - radio/TV	own - furniture/equipment	934.1	1.0	-3179.9917	

5048.1917	False		own - radio/TV	210.6139	1.0	-3853.3271
4274.5549	False		own - repairs	896.5	1.0	-3939.2275
5732.2275	False		own - vacation/others	8224.8333	0.0	2641.0162
13808.6505	True		rent - business	4083.8333	0.2346	-751.8942
8919.5608	False		rent - car	1301.2857	1.0	-2972.1197
5574.6911	False		rent - education	530.8571	1.0	-4849.8437
5911.558	False		rent - furniture/equipment	793.2857	1.0	-3557.5896
5144.161	False		rent - radio/TV	41.0	1.0	-4355.702
4437.702	False		rent - repairs	287.0	1.0	-10159.3654
10733.3654	False		free - vacation/others	6652.6667	0.8536	-4514.9677
17820.301	False		own - business	2535.9737	1.0	-7261.911
12333.8583	False		own - car	1930.4854	1.0	-7787.805
11648.7759	False		own - domestic appliances	143.5	1.0	-10302.8654
10589.8654	False		own - education	1435.0769	1.0	-8601.4663
11471.6201	False		own - furniture/equipment	1841.1	1.0	-7899.1919
11581.3919	False		own - radio/TV	1117.6139	1.0	-8601.6017
10836.8295	False		own - repairs	1803.5	1.0	-8262.8695
11869.8695	False		own - vacation/others	9131.8333	0.1818	-1314.532
19578.1987	False		rent - business	4990.8333	0.9723	-5075.5362
15057.2029	False		rent - car	2208.2857	1.0	-7600.3599
12016.9314	False		rent - education	1437.8571	1.0	-8901.3632
11777.0775	False		rent - furniture/equipment	1700.2857	1.0	-8142.3588
11542.9302	False		rent - radio/TV	948.0	1.0	-8914.9876
10810.9876	False		rent - repairs	1194.0	1.0	-12483.5029
14871.5029	False		own - business	-4116.693	0.5722	-9916.7374
1683.3514	False		own - car	-4722.1812	0.2563	-10386.7325
942.37	False		own - domestic appliances	-6509.1667	0.0852	-13347.9181
329.5848	False		own - education	-5217.5897	0.239	-11412.2787

977.0992	False	free - vacation/others	own - furniture/equipment	-4811.5667	0.236	-10513.7818
890.6485	False	free - vacation/others	own - radio/TV	-5535.0528	0.0649	-11201.1911
131.0855	False	free - vacation/others	own - repairs	-4849.1667	0.3887	-11092.064
1393.7307	False	free - vacation/others	own - vacation/others	2479.1667	0.9995	-4359.5848
9317.9181	False	free - vacation/others	rent - business	-1661.8333	1.0	-7904.7307
4581.064	False	free - vacation/others	rent - car	-4444.381	0.4224	-10262.5852
1373.8233	False	free - vacation/others	rent - education	-5214.8095	0.3769	-11888.7475
1459.1285	False	free - vacation/others	rent - furniture/equipment	-4952.381	0.2378	-10827.7212
922.9593	False	free - vacation/others	rent - radio/TV	-5704.6667	0.0736	-11614.0233
204.69	False	free - vacation/others	rent - repairs	-5458.6667	0.9761	-16626.301
5708.9677	False	own - business	own - car	-605.4882	0.9999	-2441.1428
1230.1663	False	own - business	own - domestic appliances	-2392.4737	0.9069	-6641.1215
1856.1741	False	own - business	own - education	-1100.8968	0.9996	-4208.4132
2006.6196	False	own - business	own - furniture/equipment	-694.8737	0.9996	-2643.654
1253.9066	False	own - business	own - radio/TV	-1418.3598	0.4046	-3258.9061
422.1864	False	own - business	own - repairs	-732.4737	1.0	-3935.0125
2470.0651	False	own - business	own - vacation/others	6595.8596	0.0	2347.2118
10844.5075	True	own - business	rent - business	2454.8596	0.4153	-747.6791
5657.3984	False	own - business	rent - car	-327.688	1.0	-2593.5185
1938.1426	False	own - business	rent - education	-1098.1165	1.0	-5076.0476
2879.8145	False	own - business	rent - furniture/equipment	-835.688	0.9997	-3244.4449
1573.0689	False	own - business	rent - radio/TV	-1587.9737	0.7642	-4078.5525
902.6051	False	own - business	rent - repairs	-1341.9737	1.0	-11139.8583
8455.911	False	own - car	own - domestic appliances	-1786.9854	0.9928	-5848.7134
2274.7425	False	own - car	own - education	-495.4085	1.0	-3342.0352
2351.2182	False	own - car	own - furniture/equipment	-89.3854	1.0	-1587.5076
1408.7367	False	own - car	own - radio/TV	-812.8716	0.8451	-2167.2113
541.4681	False	own - car	own - repairs	-126.9854	1.0	-3077.0499

2823.079	False					
		own - car	own - vacation/others	7201.3479	0.0	3139.62
11263.0758	True					
		own - car	rent - business	3060.3479	0.0321	110.2835
6010.4123	True					
		own - car	rent - car	277.8003	1.0	-1614.4506
2170.0512	False					
		own - car	rent - education	-492.6283	1.0	-4270.2679
3285.0113	False					
		own - car	rent - furniture/equipment	-230.1997	1.0	-2291.4458
1831.0463	False					
		own - car	rent - radio/TV	-982.4854	0.9891	-3138.7805
1173.8096	False					
		own - car	rent - repairs	-736.4854	1.0	-10454.7759
8981.805	False					
		own - domestic appliances	own - education	1291.5769	1.0	-3481.7515
6064.9053	False					
		own - domestic appliances	own - furniture/equipment	1697.6	0.9968	-2416.4917
5811.6917	False					
		own - domestic appliances	own - radio/TV	974.1139	1.0	-3089.8271
5038.0549	False					
		own - domestic appliances	own - repairs	1660.0	0.9998	-3175.7275
6495.7275	False					
		own - domestic appliances	own - vacation/others	8988.3333	0.0	3404.5162
14572.1505	True					
		own - domestic appliances	rent - business	4847.3333	0.0486	11.6058
9683.0608	True					
		own - domestic appliances	rent - car	2064.7857	0.9789	-2208.6197
6338.1911	False					
		own - domestic appliances	rent - education	1294.3571	1.0	-4086.3437
6675.058	False					
		own - domestic appliances	rent - furniture/equipment	1556.7857	0.9995	-2794.0896
5907.661	False					
		own - domestic appliances	rent - radio/TV	804.5	1.0	-3592.202
5201.202	False					
		own - domestic appliances	rent - repairs	1050.5	1.0	-9395.8654
11496.8654	False					
		own - education	own - furniture/equipment	406.0231	1.0	-2514.8331
3326.8792	False					
		own - education	own - radio/TV	-317.4631	1.0	-3167.2466
2532.3205	False					
		own - education	own - repairs	368.4231	1.0	-3503.2575
4240.1037	False					
		own - education	own - vacation/others	7696.7564	0.0	2923.428
12470.0848	True					
		own - education	rent - business	3555.7564	0.1199	-315.9242
7427.437	False					
		own - education	rent - car	773.2088	1.0	-2368.0718
3914.4894	False					
		own - education	rent - education	2.7802	1.0	-4531.2678
4536.8283	False					
		own - education	rent - furniture/equipment	265.2088	1.0	-2980.6756
3511.0932	False					
		own - education	rent - radio/TV	-487.0769	1.0	-3794.1359
2819.982	False					
		own - education	rent - repairs	-241.0769	1.0	-10277.6201

9795.4663	False				
own - furniture/equipment		own - radio/TV	-723.4861	0.9799	-2227.5981
780.6258	False				
own - furniture/equipment		own - repairs	-37.6	1.0	-3059.354
2984.154	False				
own - furniture/equipment		own - vacation/others	7290.7333	0.0	3176.6416
11404.825	True				
own - furniture/equipment		rent - business	3149.7333	0.0302	127.9794
6171.4873	True				
own - furniture/equipment		rent - car	367.1857	1.0	-1634.9957
2369.3671	False				
own - furniture/equipment		rent - education	-403.2429	1.0	-4237.1284
3430.6426	False				
own - furniture/equipment		rent - furniture/equipment	-140.8143	1.0	-2303.4174
2021.7888	False				
own - furniture/equipment		rent - radio/TV	-893.1	0.9981	-3146.4808
1360.2808	False				
own - furniture/equipment		rent - repairs	-647.1	1.0	-10387.3919
9093.1919	False				
own - radio/TV		own - repairs	685.8861	1.0	-2267.2246
3638.9968	False				
own - radio/TV		own - vacation/others	8014.2195	0.0	3950.2785
12078.1605	True				
own - radio/TV		rent - business	3873.2195	0.0006	920.1088
6826.3302	True				
own - radio/TV		rent - car	1090.6719	0.8893	-806.3248
2987.6685	False				
own - radio/TV		rent - education	320.2433	1.0	-3459.7758
4100.2623	False				
own - radio/TV		rent - furniture/equipment	582.6719	1.0	-1482.9317
2648.2754	False				
own - radio/TV		rent - radio/TV	-169.6139	1.0	-2330.0747
1990.847	False				
own - radio/TV		rent - repairs	76.3861	1.0	-9642.8295
9795.6017	False				
own - repairs		own - vacation/others	7328.3333	0.0	2492.6058
12164.0608	True				
own - repairs		rent - business	3187.3333	0.314	-761.0216
7135.6883	False				
own - repairs		rent - car	404.7857	1.0	-2830.5258
3640.0972	False				
own - repairs		rent - education	-365.6429	1.0	-4965.3372
4234.0515	False				
own - repairs		rent - furniture/equipment	-103.2143	1.0	-3440.1833
3233.7547	False				
own - repairs		rent - radio/TV	-855.5	1.0	-4252.0034
2541.0034	False				
own - repairs		rent - repairs	-609.5	1.0	-10675.8695
9456.8695	False				
own - vacation/others		rent - business	-4141.0	0.2125	-8976.7275
694.7275	False				
own - vacation/others		rent - car	-6923.5476	0.0	-11196.9531
-2650.1422	True				
own - vacation/others		rent - education	-7693.9762	0.0001	-13074.677
-2313.2753	True				
own - vacation/others		rent - furniture/equipment	-7431.5476	0.0	-11782.4229

-3080.6723	True	own - vacation/others	rent - radio/TV	-8183.8333	0.0	-12580.5354
-3787.1313	True	own - vacation/others	rent - repairs	-7937.8333	0.4331	-18384.1987
2508.532	False	rent - business	rent - car	-2782.5476	0.2058	-6017.8591
452.7639	False	rent - business	rent - education	-3552.9762	0.3999	-8152.6705
1046.7182	False	rent - business	rent - furniture/equipment	-3290.5476	0.0586	-6627.5166
46.4214	False	rent - business	rent - radio/TV	-4042.8333	0.0041	-7439.3367
-646.33	True	rent - business	rent - repairs	-3796.8333	0.999	-13863.2029
6269.5362	False	rent - car	rent - education	-770.4286	1.0	-4774.7914
3233.9342	False	rent - car	rent - furniture/equipment	-508.0	1.0	-2960.1614
1944.1614	False	rent - car	rent - radio/TV	-1260.2857	0.9712	-3792.8671
1272.2957	False	rent - car	rent - repairs	-1014.2857	1.0	-10822.9314
8794.3599	False	rent - education	rent - furniture/equipment	262.4286	1.0	-3824.5071
4349.3643	False	rent - education	rent - radio/TV	-489.8571	1.0	-4625.5453
3645.831	False	rent - education	rent - repairs	-243.8571	1.0	-10583.0775
10095.3632	False	rent - furniture/equipment	rent - radio/TV	-752.2857	1.0	-3413.5054
1908.934	False	rent - furniture/equipment	rent - repairs	-506.2857	1.0	-10348.9302
9336.3588	False	rent - radio/TV	rent - repairs	246.0	1.0	-9616.9876
10108.9876	False					

Non Parametric Test

Non-parametric tests are statistical methods used when data do not meet the assumptions required for parametric tests, such as normality and equal variances. These tests are useful for analyzing ordinal data, small sample sizes, and data with outliers or skewed distributions. Unlike parametric tests, which rely on specific probability distributions (such as the normal distribution), non-parametric methods rank the data or use distribution-free approaches.

One widely used non-parametric test is the Wilcoxon Rank-Sum Test (Mann-Whitney U Test), which compares two independent samples to determine if one tends to have higher values than the other. It is an alternative to the independent t-test when the normality assumption is violated.

For paired data, the Wilcoxon Signed-Rank Test is used instead of the paired t-test. This test ranks the absolute differences between paired observations and assesses whether the median difference is significantly different from zero.

When comparing more than two independent groups, the Kruskal-Wallis Test is the non-parametric equivalent of one-way ANOVA. It determines whether at least one group significantly differs from others by ranking the data across all groups and analyzing their distributions. If the test is significant, post-hoc tests such as Dunn's test are used for pairwise comparisons.

For repeated measures or dependent samples across multiple conditions, the Friedman Test serves as a non-parametric alternative to repeated-measures ANOVA. It ranks values within each subject and examines whether there are significant differences across conditions.

Non-parametric tests are particularly useful in real-world scenarios where data violate assumptions required for parametric methods. Though they often have less statistical power than their parametric counterparts, they provide a robust way to analyze non-normal and ordinal data.

Generate the Data set

```
In [27]: import pandas as pd
import numpy as np

# Create a sample dataset
np.random.seed(42)

df = pd.DataFrame({
    'Group': np.random.choice(['A', 'B', 'C'], size=60),
    'Score': np.random.randint(50, 100, size=60),
    'Before': np.random.randint(60, 90, size=60),
    'After': np.random.randint(65, 95, size=60)
})

#df.head() # Display first few rows
```

1. Wilcoxon Rank-Sum Test (Mann-Whitney U Test)

Used for comparing two independent samples.

```
In [28]: from scipy.stats import mannwhitneyu

group_A = df[df['Group'] == 'A']['Score']
group_B = df[df['Group'] == 'B']['Score']

stat, p = mannwhitneyu(group_A, group_B, alternative='two-sided')

print("Mann-Whitney U Test Results:")
print(f"Statistic: {stat}, P-value: {p}")
```

Mann-Whitney U Test Results:
Statistic: 101.5, P-value: 0.057541908694270685

2. Wilcoxon Signed-Rank Test

Used for paired samples (Before vs After).

```
In [29]: from scipy.stats import wilcoxon

stat, p = wilcoxon(df['Before'], df['After'])

print("Wilcoxon Signed-Rank Test Results:")
print(f"Statistic: {stat}, P-value: {p}")
```

Wilcoxon Signed-Rank Test Results:
Statistic: 375.0, P-value: 0.00019835333686059963

3. Kruskal-Wallis Test

Used for comparing more than two independent groups.

```
In [30]: from scipy.stats import kruskal

group_A = df[df['Group'] == 'A']['Score']
group_B = df[df['Group'] == 'B']['Score']
group_C = df[df['Group'] == 'C']['Score']

stat, p = kruskal(group_A, group_B, group_C)

print("Kruskal-Wallis Test Results:")
print(f"Statistic: {stat}, P-value: {p}")
```

Kruskal-Wallis Test Results:
Statistic: 4.440731200289212, P-value: 0.1085694085767165

4. Friedman Test

Used for repeated measures (Before vs After across multiple conditions).

```
In [31]: from scipy.stats import friedmanchisquare
import numpy as np
import pandas as pd

# Sample data with three time points
np.random.seed(42)
df = pd.DataFrame({
    'Before': np.random.randint(60, 90, size=30),
    'Mid': np.random.randint(65, 95, size=30),
    'After': np.random.randint(70, 100, size=30)
})
df.head()
```



```
Out[31]:
```

	Before	Mid	After
0	66	65	76
1	79	76	87
2	88	90	73
3	74	86	94
4	70	93	97

Perform Friedman Test

```
In [32]: stat, p = friedmanchisquare(df['Before'], df['Mid'], df['After'])

print("Friedman Test Results:")
print(f"Statistic: {stat}, P-value: {p}")
```

Friedman Test Results:

Statistic: 12.600000000000023, P-value: 0.0018363047770288854

Now Perform the Kruskal Wallis using German Credit Data

```
In [33]: df = pd.read_csv("german_credit__data.csv")
df.head()
```

```
Out[33]:
```

	S/no	Age	Sex	Job	Housing	Saving accounts	Checking account	Credit amount	Duration	Purpose
0	0	67	male	2	own	NaN	little	1169	6	radio/television
1	1	22	female	2	own	little	moderate	5951	48	radio/television
2	2	49	male	1	own	little	NaN	2096	12	education
3	3	45	male	2	free	little	little	7882	42	furniture/equipment
4	4	53	male	2	free	little	little	4870	24	radio/television

```
In [34]: df=df.dropna()
df.head()
```

Out[34]:

	S/no	Age	Sex	Job	Housing	Saving accounts	Checking account	Credit amount	Duration	Pi
1	1	22	female	2	own	little	moderate	5951	48	ra
3	3	45	male	2	free	little	little	7882	42	furniture/equi
4	4	53	male	2	free	little	little	4870	24	
7	7	35	male	3	rent	little	moderate	6948	36	
9	9	28	male	3	own	little	moderate	5234	30	

Grouped Summary Statistics

In [35]:

```
import pandas as pd

# Grouped summary statistics including median
summary_stats = df.groupby('Housing')['Credit amount'].agg(['count', 'mean', 'std',

# Display results
summary_stats
```

Out[35]:

	count	mean	std	min	median	max
Housing						
free	65	4846.384615	3583.997767	781	3844.0	14782
own	349	3003.504298	2770.028290	338	2136.0	18424
rent	108	3224.694444	2719.530785	276	2458.5	15945

Perform the Test

In [36]:

```
from scipy.stats import kruskal

Own = df[df['Housing'] == 'own']['Credit amount']
Free = df[df['Housing'] == 'free']['Credit amount']
Rent = df[df['Housing'] == 'rent']['Credit amount']

stat, p = kruskal(Own, Free, Rent)

print("Kruskal-Wallis Test Results:")
print(f"Statistic: {stat}, P-value: {p}")
```

Kruskal-Wallis Test Results:
Statistic: 21.06187872234066, P-value: 2.6697533871950834e-05

Homework

Practice ANOVA and Non-Parametric Tests

:

- one-way and two-way ANOVA.
- TukeyHSD for post-hoc analysis.
- the Wilcoxon rank-sum test.
- the Kruskal-Wallis test.

Homework Assignment

Exercise A: Import a Dataset

- Download a public dataset (customerData)
- Load it into R using `import()` or `read.csv()`
- Print first few rows and check structure

Exercise B: One-Way ANOVA

- Identify a numeric outcome variable and a categorical grouping variable
- Run a one-way ANOVA
- Interpret the result

Exercise C: Post-Hoc Test

- If ANOVA is significant, run Tukey's HSD
- Report which groups differ significantly

Exercise D: Two-Way ANOVA

- Include a second categorical variable

- Interpret both main effects and interaction

Exercise E: Non-Parametric Tests

- Check assumptions of ANOVA (normality, homogeneity)
- If violated, run
 - Kruskal-Wallis tests
 - Wilcoxon Signed-Rank Test
 - Perform Friedman Test
 - Wilcoxon Rank-Sum Test (Mann-Whitney U Test)
- Interpret the result

Exercise F: Write a Summary

- Summarize your findings clearly in plain language
- Include:
 - What question you were answering
 - Which variables were tested
 - Whether the result was significant
 - What conclusion can be drawn



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Thank you

